Skill

Topic 1 : Process Synchronization and Child Process Monitoring Using waitpid()

```
vennela_07@VennelaPC:~$ mkdir waitpid_project
cd waitpid_project
vennela_07@VennelaPC:~/waitpid_project$ nano waitpid_sync.c
vennela_07@VennelaPC:~/waitpid_project$ gcc -Wall -Wextra waitpid_sync.c -o waitpid_sync
vennela_07@VennelaPC:~/waitpid_project$ ./waitpid_sync
Parent process started. PID: 1491
Parent is waiting for child PID: 1492
Child process started. PID: 1492
Child is working...
Child process completed.
Child exited normally with status: 0
Parent process resumed after child completion.
```

```
  GNU nano 7.2                                                  waitpid_sync.c

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main(void) {
    pid_t pid;
    int status;

    printf("Parent process started. PID: %d\n", (int)getpid());
    fflush(stdout);

    pid = fork();

    if (pid < 0) {
        perror("fork failed");
        return 1;
    }

    if (pid == 0) {
        /* Child process */
        printf("Child process started. PID: %d\n", (int)getpid());
        printf("Child is working...\n");
        sleep(3);
        printf("Child process completed.\n");
        _exit(0);
    } else {
        /* Parent process */
        printf("Parent is waiting for child PID: %d\n", (int)pid);
        fflush(stdout);

        if (waitpid(pid, &status, 0) == -1) {
            perror("waitpid failed");
            return 1;
        }
           }

         if (WIFEXITED(status)) {
             printf("Child exited normally with status: %d\n",
                    WEXITSTATUS(status));
         } else if (WIFSIGNALED(status)) {
             printf("Child terminated by signal: %d\n",
                    WTERMSIG(status));
         }

         printf("Parent process resumed after child completion.\n");
     }

     return 0;
}
```

Topic 2 : To Retrieve PATH Variable, Parse Search Directories, Locate Executables, Verify Permissions, Handle Missing Commands, Test Command Resolution

```
vennela_07@VennelaPC:~$ mkdir path_resolution_project
cd path_resolution_project
vennela_07@VennelaPC:~/path_resolution_project$ nano path_resolution.c
vennela_07@VennelaPC:~/path_resolution_project$ gcc -Wall -Wextra path_resolution.c -o path_resolution
vennela_07@VennelaPC:~/path_resolution_project$ ./path_resolution ls
PATH variable:
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/Eclipse Adoptium/jdk-21.0.12.101-hotspot/bin:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/java8path:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/windows/system32:/mnt/c/windows:/mnt/c/windows/System32/Wbem:/mnt/c/windows/System32/WindowsPowerShell/v1.0/:/mnt/c/windows/System32/OpenSSH/:/mnt/c/Program Files/nodejs/:/mnt/c/Users/samba/AppData/Local/Programs/Python/Python314/Scripts/:/mnt/c/Users/samba/AppData/Local/Programs/Python/Python314/:/mnt/c/Users/samba/AppData/Local/Programs/Python/Launcher/:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin/:/mnt/c/Users/samba/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/samba/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/samba/AppData/Roaming/npm:/mnt/c/Users/samba/AppData/Local/Programs/DockerDesktop/resources/bin:/snap/bin

Searching for command: ls

Checking: /usr/local/sbin/ls
Checking: /usr/local/bin/ls
Checking: /usr/sbin/ls
Checking: /usr/bin/ls

Executable found: /usr/bin/ls
Permission check: Executable
vennela_07@VennelaPC:~/path_resolution_project$ ./path_resolution gcc
PATH variable:
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/Eclipse Adoptium/jdk-21.0.12.101-hotspot/bin:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/java8path:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/windows/system32:/mnt/c/windows:/mnt/c/windows/System32/Wbem:/mnt/c/windows/System32/WindowsPowerShell/v1.0/:/mnt/c/windows/System32/OpenSSH/:/mnt/c/Program Files/nodejs/:/mnt/c/Users/samba/AppData/Local/Programs/Python/Python314/Scripts/:/mnt/c/Users/samba/AppData/Local/Programs/Python/Python314/:/mnt/c/Users/samba/AppData/Local/Programs/Python/Launcher/:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin/:/mnt/c/Users/samba/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/samba/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/samba/AppData/Roaming/npm:/mnt/c/Users/samba/AppData/Local/Programs/DockerDesktop/resources/bin:/snap/bin

Searching for command: gcc

Checking: /usr/local/sbin/gcc
Checking: /usr/local/bin/gcc
Checking: /usr/sbin/gcc
Checking: /usr/bin/gcc

Executable found: /usr/bin/gcc
Permission check: Executable
vennela_07@VennelaPC:~/path_resolution_project$ ./path_resolution unknowncommand123
PATH variable:
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/Eclipse Adoptium/jdk-21.0.12.101-hotspot/bin:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/java8path:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/windows/system32:/mnt/c/windows:/mnt/c/windows/System32/Wbem:/mnt/c/windows/System32/WindowsPowerShell/v1.0/:/mnt/c/windows/System32/OpenSSH/:/mnt/c/Program Files/nodejs/:/mnt/c/Users/samba/AppData/Local/Programs/Python/Python314/Scripts/:/mnt/c/Users/samba/AppData/Local/Programs/Python/Python314/:/mnt/c/Users/samba/AppData/Local/Programs/Python/Launcher/:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin/:/mnt/c/Users/samba/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/samba/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/samba/AppData/Roaming/npm:/mnt/c/Users/samba/AppData/Local/Programs/DockerDesktop/resources/bin:/snap/bin

Searching for command: unknowncommand123

Checking: /usr/local/sbin/unknowncommand123
Checking: /usr/local/bin/unknowncommand123
Checking: /usr/sbin/unknowncommand123
Checking: /usr/bin/unknowncommand123
Checking: /sbin/unknowncommand123
Checking: /bin/unknowncommand123
Checking: /usr/games/unknowncommand123
Checking: /usr/local/games/unknowncommand123
Checking: /usr/lib/wsl/lib/unknowncommand123
Checking: /mnt/c/Program Files/Eclipse Adoptium/jdk-21.0.12.101-hotspot/bin/unknowncommand123
Checking: /mnt/c/Program Files/Common Files/Oracle/Java/javapath/unknowncommand123
Checking: /mnt/c/Program Files (x86)/Common Files/Oracle/Java/java8path/unknowncommand123
Checking: /mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath/unknowncommand123
Checking: /mnt/c/windows/system32/unknowncommand123
Checking: /mnt/c/windows/unknowncommand123
Checking: /mnt/c/windows/System32/Wbem/unknowncommand123
Checking: /mnt/c/windows/System32/WindowsPowerShell/v1.0//unknowncommand123
Checking: /mnt/c/windows/System32/OpenSSH//unknowncommand123
Checking: /mnt/c/Program Files/nodejs//unknowncommand123
Checking: /mnt/c/Users/samba/AppData/Local/Programs/Python/Python314/Scripts//unknowncommand123
Checking: /mnt/c/Users/samba/AppData/Local/Programs/Python/Python314//unknowncommand123
Checking: /mnt/c/Users/samba/AppData/Local/Programs/Python/Launcher//unknowncommand123
Checking: /mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin//unknowncommand123
Checking: /mnt/c/Users/samba/AppData/Local/Microsoft/WindowsApps/unknowncommand123
Checking: /mnt/c/Users/samba/AppData/Local/Programs/Microsoft VS Code/bin/unknowncommand123
Checking: /mnt/c/Users/samba/AppData/Roaming/npm/unknowncommand123
Checking: /mnt/c/Users/samba/AppData/Local/Programs/DockerDesktop/resources/bin/unknowncommand123
Checking: /snap/bin/unknowncommand123
```

```
Command not found or no executable permission: unknowncommand123
vennela_07@VennelaPC:~/path_resolution_project$ ./path_resolution /usr/bin/ls
Executable found: /usr/bin/ls
Permission check: Executable
```

GNU nano 7.2 path_resolution.c

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/stat.h>
#include <limits.h>

#ifndef PATH_MAX
#define PATH_MAX 4096
#endif

int main(int argc, char *argv[]) {
    const char *path;
    char *path_copy;
    char *dir;
    char *saveptr = NULL;
    char full_path[PATH_MAX];
    struct stat info;
    int found = 0;

    if (argc != 2) {
        printf("Usage: %s command\n", argv[0]);
        return 1;
    }

    /* Commands containing / are checked directly */
    if (strchr(argv[1], '/') != NULL) {
        if (stat(argv[1], &info) == 0 &&
            S_ISREG(info.st_mode) &&
            access(argv[1], X_OK) == 0) {
            printf("Executable found: %s\n", argv[1]);
            printf("Permission check: Executable\n");
            return 0;
        }

        perror("Command not found or not executable");
        return 1;
    }

    path = getenv("PATH");

    if (path == NULL) {
        printf("PATH variable is not set.\n");
        return 1;
    }

    printf("PATH variable:\n%s\n\n", path);

    path_copy = malloc(strlen(path) + 1);

    if (path_copy == NULL) {
        perror("malloc failed");
        return 1;
    }

    strcpy(path_copy, path);

    printf("Searching for command: %s\n\n", argv[1]);

    /* Parse PATH directories */
    dir = strtok_r(path_copy, ":", &saveptr);
```

```c
    while (dir != NULL) {
        /* Empty PATH entries represent the current directory */
        const char *search_dir = (*dir == '\0') ? "." : dir;
        int result = snprintf(full_path, sizeof(full_path),
                              "%s/%s", search_dir, argv[1]);

        if (result < 0 || (size_t)result >= sizeof(full_path)) {
            printf("Path is too long; skipping directory.\n");
            dir = strtok_r(NULL, ":", &saveptr);
            continue;
        }

        printf("Checking: %s\n", full_path);

        if (stat(full_path, &info) == 0 &&
            S_ISREG(info.st_mode)) {
            if (access(full_path, X_OK) == 0) {
                printf("\nExecutable found: %s\n", full_path);
                printf("Permission check: Executable\n");
                found = 1;
                break;
            } else {
                printf("File exists but is not executable.\n");
            }
        }

        dir = strtok_r(NULL, ":", &saveptr);
    }

    if (!found) {
        printf("\nCommand not found or no executable permission: %s\n",
               argv[1]);
    }

    free(path_copy);
    return found ? 0 : 1;
}
```

Topic 3 : To Detect Variable References, Expand Values, Handle Undefined Variables, Update Tokens, Support Nested Variables, Test Expansion Logic

```
vennela_07@VennelaPC:~/path_resolution_project$ mkdir variable_expansion_project
cd variable_expansion_project
vennela_07@VennelaPC:~/path_resolution_project/variable_expansion_project$ nano variable_expansion.c
vennela_07@VennelaPC:~/path_resolution_project/variable_expansion_project$ gcc -Wall -Wextra variable_expansion.c -o variable_expansion
vennela_07@VennelaPC:~/path_resolution_project/variable_expansion_project$ ./variable_expansion
Variable Expansion Parser
Examples: $HOME, ${USER_NAME}, $MESSAGE, ${$INNER}
Type exit to quit.

Enter text: $HOME
Original input: $HOME
Expanded output: /home/vennela_07
Validation successful.

Enter text: ${USER_NAME}
Original input: ${USER_NAME}
Expanded output: Vennela
Validation successful.

Enter text: $MESSAGE
Original input: $MESSAGE
Expanded output: Hello Vennela
Validation successful.

Enter text: ${$INNER}
Original input: ${$INNER}
Expanded output: Vennela
Validation successful.
```

```
Enter text: Hello $UNKNOWN_VARIABLE
Original input: Hello $UNKNOWN_VARIABLE
Expanded output: Hello
Validation successful.

Enter text: ${HOME
Error: Missing closing brace.
Error: Expansion failed or output limit exceeded.

Enter text: exit
```

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <ctype.h>

#define INPUT_SIZE 512
#define OUTPUT_SIZE 2048
#define MAX_DEPTH 10

int expand_variables(const char *input, char *output,
                     size_t output_size, int depth) {
    size_t i = 0, out_len = 0;

    if (depth > MAX_DEPTH) {
        fprintf(stderr, "Error: Maximum expansion depth exceeded.\n");
        return 0;
    }

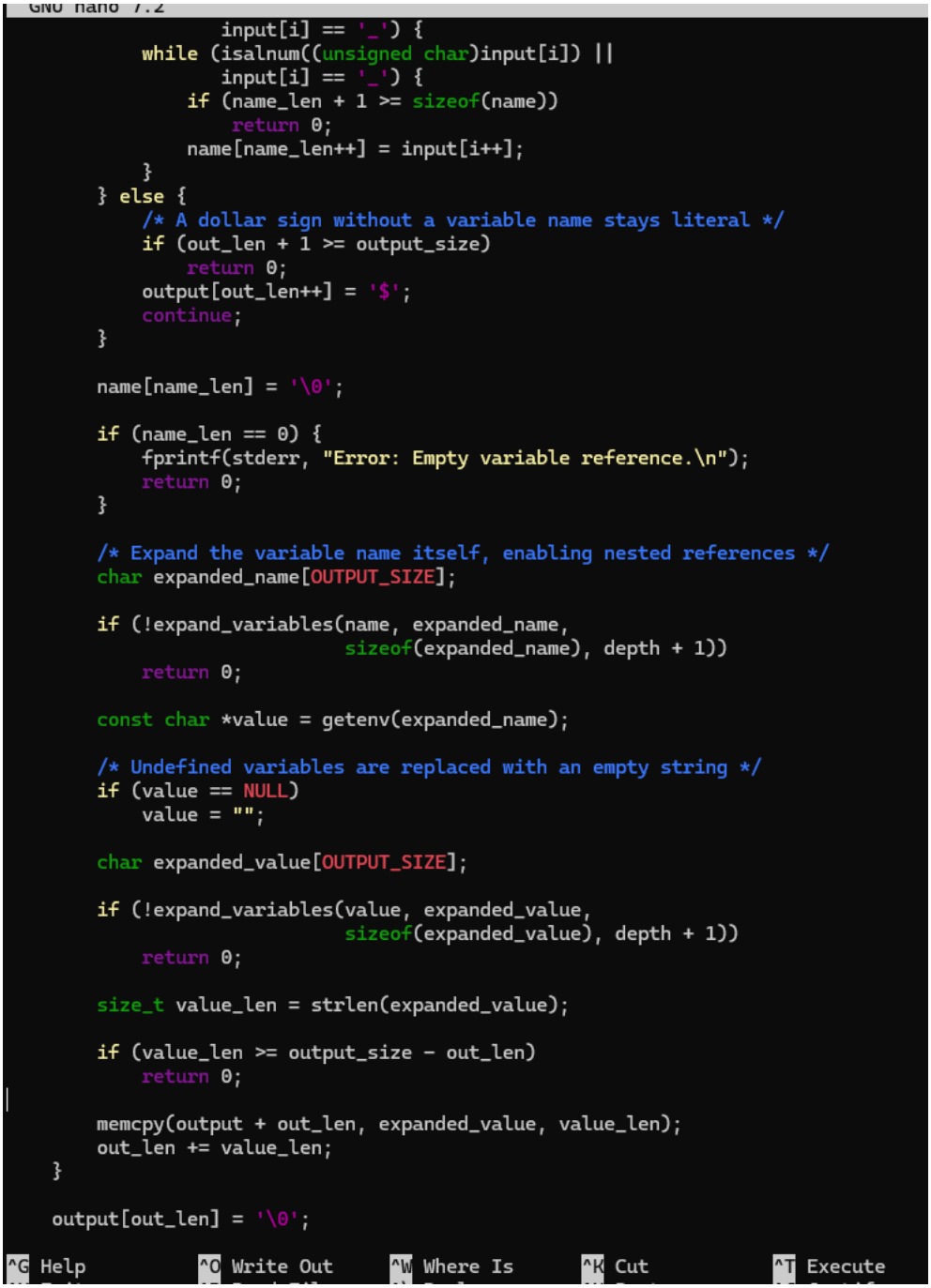
    while (input[i] != '\0') {
        if (input[i] != '$') {
            if (out_len + 1 >= output_size)
                return 0;

            output[out_len++] = input[i++];
            continue;
        }

        i++;
        char name[256];
        size_t name_len = 0;

        if (input[i] == '{') {
            i++;
            while (input[i] != '\0' && input[i] != '}') {
                if (name_len + 1 >= sizeof(name))
                    return 0;
                name[name_len++] = input[i++];
            }

            if (input[i] != '}') {
                fprintf(stderr, "Error: Missing closing brace.\n");
                return 0;
            }
            i++;
        } else if (isalpha((unsigned char)input[i]) ||
```


GNU nano 7.2
input[i] == '_') {
while (isalnum((unsigned char)input[i]) ||
input[i] == '_') {
if (name_len + 1 >= sizeof(name))
return 0;
name[name_len++] = input[i++];
}
} else {
/* A dollar sign without a variable name stays literal */
if (out_len + 1 >= output_size)
return 0;
output[out_len++] = '$';
continue;
}
name[name_len] = '\0';
if (name_len == 0) {
fprintf(stderr, "Error: Empty variable reference.\n");
return 0;
}
/* Expand the variable name itself, enabling nested references */
char expanded_name[OUTPUT_SIZE];
if (!expand_variables(name, expanded_name,
sizeof(expanded_name), depth + 1))
return 0;
const char *value = getenv(expanded_name);
/* Undefined variables are replaced with an empty string */
if (value == NULL)
value = "";
char expanded_value[OUTPUT_SIZE];
if (!expand_variables(value, expanded_value,
sizeof(expanded_value), depth + 1))
return 0;
size_t value_len = strlen(expanded_value);
if (value_len >= output_size - out_len)
return 0;
memcpy(output + out_len, expanded_value, value_len);
out_len += value_len;
}
output[out_len] = '\0';
^G Help
^O Write Out
^W Where Is
^K Cut
^T Execute

```c
}

int main(void) {
    char input[INPUT_SIZE];
    char output[OUTPUT_SIZE];

    /* Demo variables for testing nested expansion */
    setenv("USER_NAME", "Vennela", 1);
    setenv("GREETING", "Hello", 1);
    setenv("MESSAGE", "$GREETING $USER_NAME", 1);
    setenv("INNER", "USER_NAME", 1);

    printf("Variable Expansion Parser\n");
    printf("Examples: $HOME, ${USER_NAME}, $MESSAGE, ${$INNER}\n");
    printf("Type exit to quit.\n");

    while (1) {
        printf("\nEnter text: ");

        if (fgets(input, sizeof(input), stdin) == NULL)
            break;

        if (strchr(input, '\n') == NULL && !feof(stdin)) {
            int ch;
            while ((ch = getchar()) != '\n' && ch != EOF) {}
            printf("Error: Input too long.\n");
            continue;
        }

        input[strcspn(input, "\n")] = '\0';

        if (strcmp(input, "exit") == 0)
            break;

        if (expand_variables(input, output, sizeof(output), 0)) {
            printf("Original input: %s\n", input);
            printf("Expanded output: %s\n", output);
            printf("Validation successful.\n");
        } else {
            printf("Error: Expansion failed or output limit exceeded.\n");
        }
    }

    printf("Parser terminated.\n");
    return 0;
}
```

Topic 4 : To Identify Built-ins, Create Dispatch Table, Execute In-Process Commands, Handle Invalid Commands, Maintain State, Test Dispatch Logic

```
vennela_07@VennelaPC:~$ mkdir builtin_dispatch_project
cd builtin_dispatch_project
vennela_07@VennelaPC:~/builtin_dispatch_project$ nano builtin_dispatch.c
vennela_07@VennelaPC:~/builtin_dispatch_project$ gcc -Wall -Wextra builtin_dispatch.c -o builtin_dispatch
vennela_07@VennelaPC:~/builtin_dispatch_project$ ./builtin_dispatch
Built-in Command Shell
Type 'help' for commands.
shell> help
Available built-in commands:
help  - Display available commands
cd    - Change current directory
pwd   - Display current directory
set   - Set a shell variable
get   - Display a shell variable
exit  - Exit the shell
```

```
shell> pwd
/home/vennela_07/builtin_dispatch_project
shell> cd ..
pwdshell>
/home/vennela_07
shell> set COURSE OS
geVariable COURSE set successfully.
shell> t COURSE
COURSE=OS
shell> hello
Invalid command: hello
Type 'help' to see available commands.
shell> exit
Shell terminated.
```

GNU nano 7.2 builtin_dispatch.c

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <limits.h>

#define INPUT_SIZE 512
#define MAX_ARGS 20

typedef int (*CommandFunction)(char **args);

typedef struct {
    const char *name;
    CommandFunction function;
} BuiltinCommand;

/* Built-in command functions */

int command_help(char **args) {
    (void)args;

    printf("Available built-in commands:\n");
    printf("help  - Display available commands\n");
    printf("cd    - Change current directory\n");
    printf("pwd   - Display current directory\n");
    printf("set   - Set a shell variable\n");
    printf("get   - Display a shell variable\n");
    printf("exit  - Exit the shell\n");

    return 0;
}
```